

LIN7216 – Formal Linguistics

Constructing and interpreting simple sentences

Adèle Hénot-Mortier

24/09/2026

Queen Mary University of London

- Adèle Hénot-Mortier (she/her).
- Thursday 9-11 , Bancroft 3.27.¹
- Office hours Monday 2-3 and Wednesday 10-11 in my office (ArtsOne 1.13) or on Teams.
- [QMPlus](#).
- [GitHub](#) (lecture notes only).

¹An additional weekly 1h session will be scheduled starting next week.

Course overview

- **Thematic structure:** each week starting next week we'll focus on a different linguistic object, and analyze it from a formal angle, specifically structure (syntax) and meaning (semantics).

- We will try to start sessions with a short **brainstorming phase**



- during which we will share, list and organize all we already know about the linguistic object(s) at stake.
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- QMPlus includes **weekly readings**. Core readings (textbook chapters mostly) are **strongly recommended** and directly complement the in-person sessions, while advanced readings (academic papers) are **optional**.
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Assessments

- Two **identical assessments**: data analysis writeups.
 - Assessment 1 (~ 3000 words) due Wednesday, **November 4th** (middle of reading week).
 - Assessment 2 (~ 3000 words) due Wednesday, **December 11th** (week 12).
- The writeups will be based on the Week 5 and Week 9 sessions.
- During these sessions, we will **analyze unfamiliar linguistic data** together and try to come up with an analysis.
- You will summarize (and potentially improve) the analysis in your assessments, using the **conventions of a linguistics journal paper**: examples, glosses, meaningful sections, scientific



approach, clear argumentation

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Use of AI

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Brainstorming

The core elements of a sentence

- What's a “good” sentence?
- What typically makes up a sentence?
- How are the elements organized?
- Can you come up with the longest sentence that exists?

Syntax as a data-driven science

- Let's rate the following sentences 🤔

- (1)
 - a. Big black cats purr loudly.
 - b. Colorless green ideas sleep furiously.
 - c. Furiously sleep ideas green colorless.
 - d. Who did you give this book to?
 - e. She (be) working late.
 - f. The car needs washed.
 - g. Who do you think's cat is cute?
 - h. Which problem do you wonder how she can solve?

Grammaticality vs. standard vs. felicity

- As **syntacticians**, we are interested in what is **well-formed**, quite independently of assigned meaning and plausibility.
- Well-formed does not mean **standard**! It means a structure that could be uttered spontaneously in a natural setting, by native speakers of a given language (variety). This corresponds to a **descriptivist** (i.e. non-prescriptivist) approach.
- Well-formed does not mean **probable** either.² Some oddly constructed/convoluted sentences can still be recognized as part of the language, even when nobody has uttered them before.
- Sentences that are well-formed but do not have a “good” meaning in context are **infelicitous**. Felicity is a **pragmatic** concept.

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- Linguists use **diacritics** to categorize sentences.
 - Ill-formed/ungrammatical sentences are marked *;
 - borderline-grammatical sentences ?, and sentences whose acceptability depends on speakers and/or language variety are %.
 - Infelicitous sentences are #.

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| | c. | * Furiously sleep ideas green colorless. | Ill-formed |
| | d. | Who did you give this book to? | Well-formed, not "literary" |
| | e. | % ³ She (be) working late. | Well-formed in AAE |
| | f. | % ⁴ The car needs washed. | Well-formed in e.g. Scotland |
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 - Predicts the grammatical sentences to be grammatical.
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 - Has the **parametric flexibility** to explain the variable cases, language-internally (dialectal variation, %), and externally (cross-linguistic variation).
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The building blocks of sentences

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- And constituents can also be grouped with other constituents to form even larger constituents!
- For instance, in the sentence *The big cat and the small dog are playing with the ball*, which words seem to form intuitive chunks?

Forming constituents with Merge

- The operation of putting words/constituents together is called **MERGE** in the Minimalist framework.
- Let's try to inductively merge the elements of (2), starting with single words. We'll write two merged (adjacent) elements X and Y $[XY]$.

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The big cat is playing with the ball and the small dog ~~is playing with the ball~~ too.

Syntax trees

Drawing trees

- Below is a parse of the previous sentence, together with partial labeling.⁶
- (2) [[[_{DP}The [_{NP}big cat]] [and [_{DP}the [_{NP}small dog]]]] [are [_{VP}playing [_{PP}with [_{DP}the ball]]]]].
- The [.] notation used previously to signal merged elements (constituents) is compact but quite heavy 😞
 - Instead, linguists often use trees. If X and Y merge together to form [_{XP} X Y] (where the head X projects), [_{XP} X Y] can be represented as an (unordered) binary branching tree rooted in XP and with daughters X and Y :



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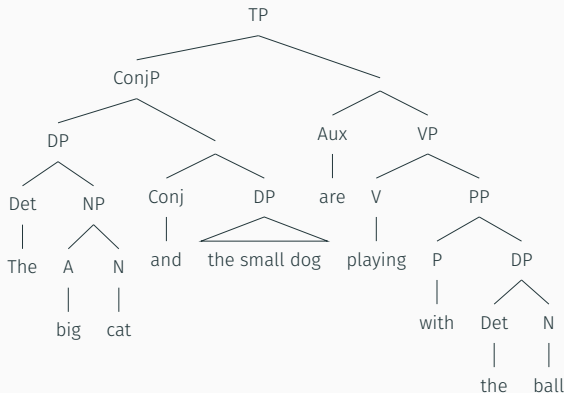
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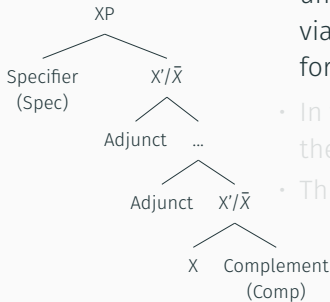
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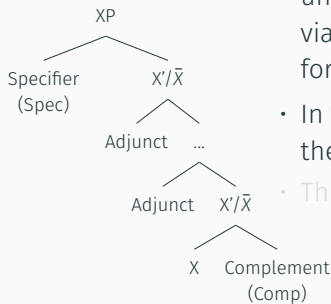
- We use triangles to abstract over obvious/irrelevant structure.
- We consider full standalone sentences (**matrix sentences**) like (2) to be Tense Phrases (TPs).

The X-bar schema



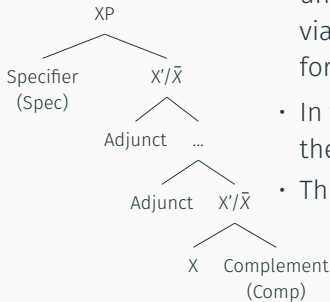
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- In the X-bar schema any phrase *XP* follows the same recursive “template” on the left.
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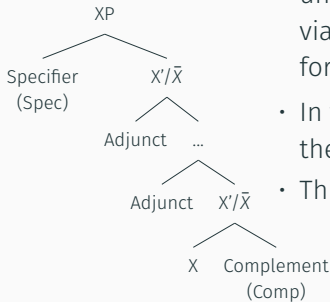
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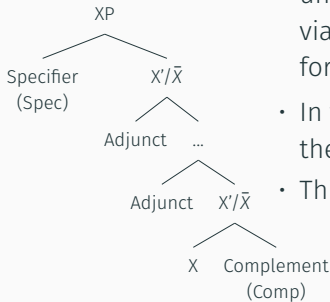
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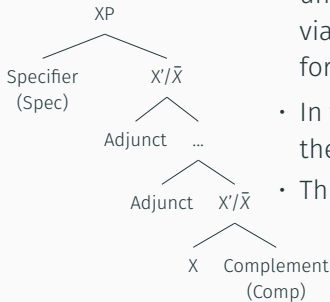
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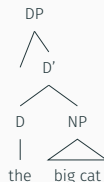
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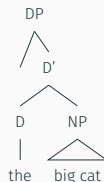
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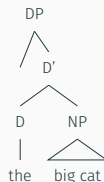
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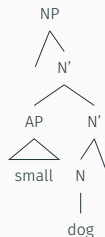
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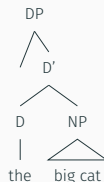


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Head is *dog* (N), no Spec, one adjunct (the AP *small*), no Comp.

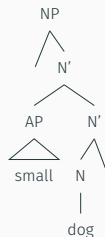


Applying the X-bar schema

- In *the big cat*, what's the head? Is there a specifier? An adjunct? A complement?
Head is *the* (D), no Spec, Comp is the NP *big cat*.



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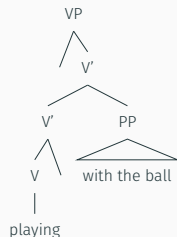


Applying the X-bar schema cont'd

- In *playing with the ball*, what's the head? Is there a specifier? An adjunct? A complement?

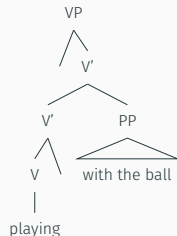
Applying the X-bar schema cont'd

- In *playing with the ball*, what's the head? Is there a specifier? An adjunct? A complement?
Head is *playing* (V), no Spec, one adjunct (arguably, the PP *with the ball*), no Comp.



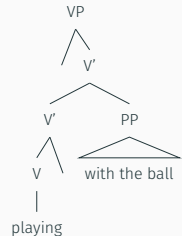
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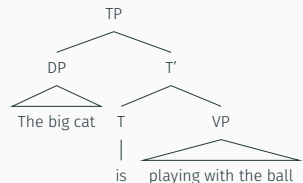


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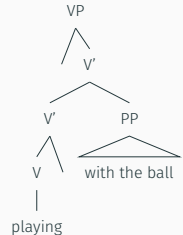


- In *The big cat is playing with the ball*, what's the head? Is there a specifier? An adjunct? A complement?
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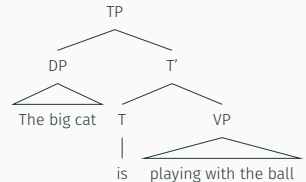


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- **For next week:** draw the complete tree for *The big cat and the small dog are playing with the ball* using X-bar theory.

More X-bar vocabulary

- X' (pronounce X-bar) are called **intermediate projections**, XP **maximal projection**.
- A phrase is a subtree rooted in an XP.
- A constituent is a subtree rooted in an XP, X', or a head X.
- Is every phrase a constituent? Every constituent a phrase?
- Assuming auxiliaries are in T, *will leave* is a non-phrasal constituent (rooted in T'), which for instance, can undergo ellipsis:

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More on projections

- **Lexical** projections like NP or VP have meaning highly dependent on which lexical item (open-class) instantiates the head (compare *jump/give*, or *cat/freedom*).
- **Functional** projections like DP, PP have more deterministic, “functional” meanings, picked from a finite set (closed class, e.g. *a/the*, and *to/from*).
- **Extended** projections are sequences of projections embedding each other. The lowest one is lexical, while the higher ones are functional. PP-DP-NumP⁷-NP is a typical extended projection in

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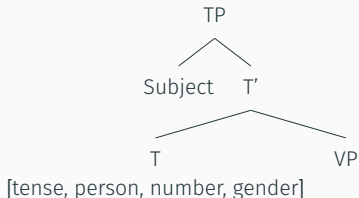
Why is *the cat* a DP and not an NP?

- Whether *the cat* should be treated as a DP (headed by *the*) or an NP (headed by *cat*) is still a matter of heated debate, but **the dominant view is the DP view**. Some evidence for this:
 - Ds **cannot be iterated** (which would be possible assuming they were adjuncts of N): **The this cat*.
 - It makes sense to analyze pronouns like *you* (personal), *this*, *that* (demonstrative) as D-heads with a null complement, because both pronouns and determiners are functional, closed-class elements. And because **pronouns distribute like Det-N sequences**, Det-N is best analyzed as headed by D.
 - It makes sense to analyze **possessors** in Saxon genitives like *Jo's book* as specifiers of null determiners, with a possessee complement: $[_{DP}[Jo's][_{D'}[_{D}\emptyset][_{NP}book]]]$.
 - More theory-internally, having functional elements (like Ds) embed lexical elements (like Ns) is **in line with the concept of extended projections**, supported elsewhere in the grammar.⁸

⁸Another example on next slide.

Why are sentences TPs?

- Older frameworks assigned sentences a label S, which was not projected from any of the daughter phrases (typically, an NP subject and a VP). This is **problematic for X-bar theory**.
- The TP solution⁹ does not “choose” between the N and V labels and instead introduces another projection (TP):
 - whose head stores tense/inflectional information (functional information);
 - whose specifier hosts the subject;
 - whose complement is the VP.



⁹You might see “IP” (Inflection Phrase) instead in certain textbooks—roughly the same thing, except IP is more of an umbrella term.

Why are sentences TPs? cont'd

- Treating subjects TP-specifiers is in line with the idea that subjects, *via* **agreement** with the tensed auxiliary or main verb, give the sentence its featural makeup (person, gender, number).
- There's also evidence that sentences "behave" like tensed objects: tense (finite/non-finite) is a feature that clause-embedding verbs are sensitive to. Some select for finite, some for non-finite.

- (4) a. Jo remembers that Al **left**. Finite embedding
 b. Jo arranged for Al **to leave**. Infinitival embedding

- Also notice the parallel with the DP(NP) structure: in a TP(VP) structure, a functional layer (T), embeds a lexical layer (V), to form a (partial¹⁰) extended projection.

¹⁰We'll see that clauses can contain a lot of extra outer functional structure, e.g. CP.

Generative capacity

Infinite outputs from finite means

- We just defined a structure-building system out of a **finite set of labels** (N, D, V, T...), instantiated by a **finite set of lexical items** (*cat, the, purr...*), and following a **fixed tree schema**. The parameters/inputs to this model are finite in number.

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 - Second (stronger argument), clauses can be **embedded *ad infinitum***: *Jo thinks that Al believes that Sue knows [...] that Ed has left*.
- This comes from the recursivity of the X-bar schema and is important from a cognitive perspective: the finite set of parameters can be stored in the brain as a **Universal Grammar (UG)**, and acquired fairly fast, with the capacity to match the infinity of our outputs 🧠.

Inifinite outputs, really?

- One could argue the sentences used to motivate the intinity of our outputs get extremely long, and thus unpronouncable/impossible to fully remember!
- It seems to be more about our **general cognitive limitations**, not tied to our internal grammar (UG): working memory, cognitive load, limitations or the articulatory system...
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Finite means, really?

- What about **open-class categories**? Our lexicon is continually enriched with new nouns, adjectives and verbs, so can we really say our building blocks form a finite set?
- First, there are reasons to think word formation (morphology) follows rules very similar to syntactic rules. In fact, some people argue **morphology is virtually syntax**.
- For instance, one can form a tree for a word like *un-character-istic-al-ly*, which contains a root (*character*), a prefix (*un-*) and multiple suffixes. This **reduces the number of lexical items to posit**, as affixes are a closed-class, and roots often get reused in different words.
- But more fundamentally, the “finite means” of UG do not refer to all the building blocks (roots, words) that *could* be there. Instead, UG ensures that **from any lexicon of fixed, finite size, one can get an infinite number of sentences**.

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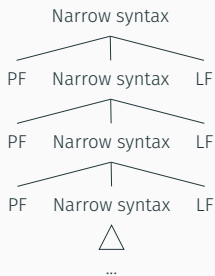
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Interpeting trees

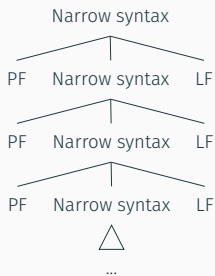
The Minimalist model (aka the Squished Catterpillar¹¹)



- As structure gets built, it gets sent for actual **production** to phonology (PF) and for **interpretation** to semantics (LF) at regular “checkpoints”, called **phases**.
- PF and LF are also called interfaces. Minimalist syntax is often called “narrow” because it delegates a lot of computational work to the interfaces.
- We will now focus on the LF interface, which assigns meaning to trees.

¹¹©Norvin Richards

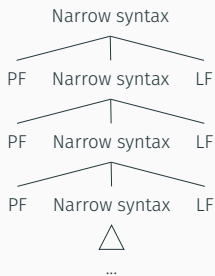
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Meaning as truth conditions

- The meaning of a sentence are the **conditions under which it is true**. This is the assumption made by **Extensional Semantics**, that we will be using for most of the semester.
- Sentence meaning is evaluated against a situation, materialized as a short **scenario** in linguistic examples. If the scenario satisfies the sentence's truth conditions, the sentence evaluates to the truth-value 1 (and we may flag it as ✓), and if not, the sentence evaluates to 0 (flagged as ✗).
- Some sentence may feel “weird”/uninterpretable in context, but not clearly false. Those will be flagged with #.
 - (5) Scenario: *Jo is smiling and Al is crying.*
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- Just like syntacticians, semanticists use (truth-value) judgments to find the logical interpretation of a sentence that evaluates to 1 exactly in the scenarios where the sentence sounds true. Like syntax, **the ideal semantic system should appeal to a finite set of rules and primitive meanings.**
- We will design a **compositional** system that has these properties: the meaning of a sentence will be derived from the meaning of its individual constituents and how they (locally) combine together.
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Simplest model: individuals and one-place predicates

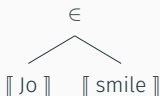
- We can build a simple semantic formalism (**metalanguage**) based on **set theory**.
- We posit **individuals** and **sets of individuals** as primitive types of the metalanguage.
- A proper name like *Jo* (a syntactic object from the **object language** that is being interpreted) can be taken to denote the *individual* *Jo*, noted *j* (metalanguage). We write $\llbracket Jo \rrbracket = j$ (“the denotation of the word *Jo* is the individual *j*”).
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Deriving the meaning of simple sentences

- A sentence like *Jo smiles* can then be derived by interpreting its two words, and assuming that they semantically combine via set membership relation.



$$\begin{aligned}\llbracket Jo \text{ smile} \rrbracket &= 1 \text{ iff } \llbracket Jo \rrbracket \in \llbracket smile \rrbracket \\ &= 1 \text{ iff } j \in \{x | x \text{ smiles} \} \\ &= 1 \text{ iff } j \text{ smiles}\end{aligned}$$

- We obtained an interpretation of a sentence from the object language (*Jo smiles*) as a truth condition in the metalanguage (true if and only if the individual *Jo* is a smiling individual).



- This may not seem like much, but shows that from simple lexical entries like $\llbracket Jo \rrbracket = j$ one can *compute* the truth conditions of entire sentences.
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Foreshadowing next week: functional meaning

- Suppose negation (expressed by *not* or the prefix *un-*) is a function taking a predicate (set of individuals) and returning its complement set, i.e. all the individuals that are **not** in the original set: $\llbracket \text{not} \rrbracket = \llbracket \text{un-} \rrbracket = S \mapsto \bar{S}$ (“the function mapping any set S to its complement set”).
- One can then compute the denotation of the predicates *not happy/un-happy*, assuming branching semantically amounts to applying a function (negation) to its argument (*happy*).

Function Application


$$\begin{aligned}\llbracket \text{not happy} \rrbracket &= \llbracket \text{un-happy} \rrbracket = \llbracket \text{not} \rrbracket(\llbracket \text{happy} \rrbracket) \\ &= \llbracket \text{not} \rrbracket(\{x \mid x \text{ is happy}\}) \\ &= \{x \mid x \text{ is not happy}\}\end{aligned}$$

$\llbracket \text{not/un-} \rrbracket$ $\llbracket \text{happy} \rrbracket$

- This shows semantics can sometimes apply similarly **above and below the word**.
- Food for thought: reflect on the meanings of *unlockable*, and *unhappier*. Any puzzle/issue?

Foreshadowing next week: functional meaning

- Suppose negation (expressed by *not* or the prefix *un-*) is a function taking a predicate (set of individuals) and returning its complement set, i.e. all the individuals that are **not** in the original set: $\llbracket \text{not} \rrbracket = \llbracket \text{un-} \rrbracket = S \mapsto \bar{S}$ (“the function mapping any set S to its complement set”).
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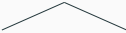
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